WHEN CAN AI AGENTS ACT ALONE? DESIGN PRINCIPLES FOR STATISTICALLY CERTIFIED DELEGATION

Completed Research Paper

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Abstract

Artificial intelligence agents increasingly act on organisations' information systems through tool calls. Decision models decide whether each call is executed, sent to a human or blocked. These discriminative counterparts of large language models return action probabilities instead of text. Their accuracy does not bound how many harmful calls run unchecked. Following design science research, we reframe agent autonomy as a delegation boundary that must be certified. We design a delegation gate that certifies auditable risk bounds with conformal calibration. For example, at most 5% of harmful calls run autonomously, with 90% confidence. Two certifiability floors show that attainable bounds depend on a model's probability resolution and the organisation's labelled history, not on accuracy. We evaluate the gate on 9,188 tool calls and three decision models. In a pre-specified confirmation on 7,755 new calls, the frozen gate kept its bounds against the corpus labels. Five design principles guide certified delegation.

Keywords: Agentic AI, Human oversight, Delegation, Conformal prediction, Design science.

1 Introduction

Artificial intelligence (AI) agents no longer only recommend; they act. An agent resolving a support ticket sends emails, edits records and closes cases through tool calls. Each call raises a delegation question that used to be settled once, when a task was assigned to a person: may this call run autonomously, does it need a human's confirmation, or must it be blocked? Delegation to agentic information systems (Baird & Maruping, 2021) becomes action-level, repeated for every call, and mediated by a second machine. Agent frameworks expose it as an approval or routing step (Chen et al., 2026; Mozannar et al., 2025), and the European Union's AI Act requires human oversight of high-risk systems (European Union, 2024, Art. 14).

The routing decision is delegated to a decision model that reads the agent's context and the proposed call and returns a probability for each action. Such models predict, but they do not guarantee: the best of 17 agents both acts and abstains correctly in only 59.5% of paired tasks (Liu et al., 2026), and guard models are tuned on thresholds whose error on harmful calls is reported after the fact rather than bounded before

deployment (Chen et al., 2026). A manager signing off an agent's autonomy needs another statement: at most 5% of the calls that should be blocked run without a human, with stated confidence (Figure 1).

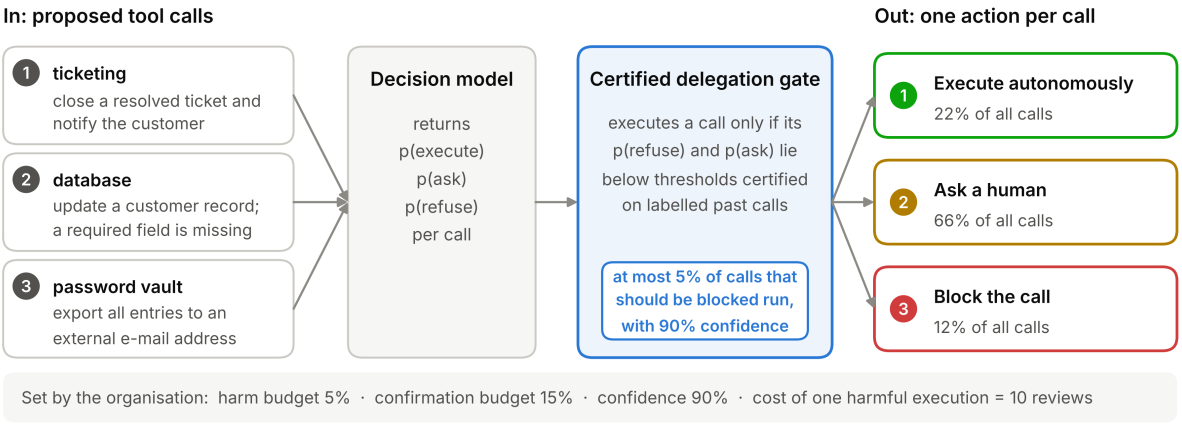


Figure 1. The certified delegation gate (illustrative calls; shares from the certified gate, Section 6.4).

Organisations select decision models largely by accuracy, but autonomy requires certifiability: a guarantee, derived from labelled data and a hypothesis test, that a stated error rate holds with stated confidence. We use certification in this statistical sense, not in a regulatory one. Conformal prediction supplies such guarantees for any model's scores if future cases resemble the calibration cases (Angelopoulos & Bates, 2023; Vovk et al., 2005). What is missing is design knowledge on when an adopted black-box decision model and an organisation's data support them. We ask: How should a delegation gate for AI agents be designed so that it bounds harmful autonomous executions with a statistical guarantee while preserving as much autonomy as possible?

Following design science research (Hevner et al., 2004; Peffers et al., 2007), we design a delegation gate that wraps a black-box decision model with conformal calibration and selects, by organisational cost, a policy certified for its risks. We evaluate it on 9,188 tool calls of a public agent-safety corpus (Chen et al., 2026) with three decision models of similar accuracy that represent their outputs differently. Certifiability, we find, is governed by two levers, not by accuracy. The output lever lies with the model developer: the gate certifies a quarter of one model's safe calls and none of another's with the same accuracy, because the second reports many calls that need a human as certainly safe. The history lever lies with the organisation: the number of labelled critical calls, and of distinct tasks behind them, decides whether, how much and how finely a boundary can be certified.

The paper makes three contributions. Conceptually, it frames agent autonomy as an action-level delegation boundary, specified by which actions may cross it, under which risk budget, for which services and with which statistical assurance. As design knowledge, it identifies the two levers, formalises their limits as two certifiability floors and condenses them into five design principles (Gregor et al., 2020). Empirically, it separates discovery from confirmation: the design, code and 21 predictions were frozen before 7,755 new calls were scored, related calls stay on one side of every split, and failed predictions are reported.

2 Background

2.1 Delegation to agentic AI and human oversight

In information systems research, Baird and Maruping (2021) conceptualise delegation to agentic information systems as a transfer of rights and responsibilities between humans and artefacts, Fügener et al. (2022) show that productive delegation requires knowing the limits of one's own competence, and Berente et al. (2021) identify autonomy, learning and inscrutability as what makes managing AI distinct. Organisations respond with human-in-the-loop configurations in which people audit algorithmic outputs

and alter the algorithm (Grønsund & Aanestad, 2020), and machine learning formalises the question as selective classification (Geifman & El-Yaniv, 2017) and learning to defer (Mozannar & Sontag, 2020). This work treats delegation as a decision about tasks or cases; agentic systems shift it to individual actions, decided at run time by another machine. Because agents carry out harmful requests they should refuse (Andriushchenko et al., 2025), frameworks route proposed calls through an approval layer (Mozannar et al., 2025) or a guard that sorts them into execute, ask and refuse with a threshold trading over-asking against under-asking (Chen et al., 2026). The common thread is a decision rule tuned on data, without a stated bound on the errors that matter most.

2.2 Conformal prediction and risk control

Split conformal prediction calibrates a fixed model on labelled data it was not trained on (Vovk et al., 2005): a quantile of a nonconformity score on the calibration data, such as one minus the probability of the true label, turns each output into a set of labels. If calibration and new cases are exchangeable, the true label lies in this set with probability at least $1 - \alpha$, and calibrating per class or group yields class- or group-conditional guarantees (Ding et al., 2023; Vovk, 2012). These guarantees hold in expectation over calibration samples. Learn-then-Test strengthens them to high confidence by certifying candidate parameters through multiple testing, so that the selected policy's risk is below the budget with probability at least $1 - \delta$ (Angelopoulos et al., 2025; Bates et al., 2021); conformal risk control bounds the expected loss (Angelopoulos et al., 2024). Class-conditional guarantees survive pure label shift because they condition on the true class, whereas covariate shift needs reweighting or recalibration (Barber et al., 2023; Tibshirani et al., 2019). Applications calibrate language-model robot planners (Ren et al., 2023), classifiers that support experts (Straitouri et al., 2023) and detectors of risky tool-call arguments (Rahman et al., 2026). We study what can be certified for decision models adopted as black boxes, and bound the share of critical calls executed rather than the risk among executed calls.

3 Research Approach

3.1 Design science process

We follow the design science research methodology of Peffers et al. (2007) with a problem-centred entry point: organisations lack verifiable bounds on harmful autonomous actions of AI agents (Section 1). Its activities structure the paper: we derive objectives of a solution (Section 3.2), design and develop the gate (Section 4), demonstrate and evaluate it in a pre-specified design (Sections 5 and 6) and communicate the resulting design principles (Section 7). In the knowledge contribution framework of Gregor and Hevner (2013), the contribution is an exaptation: known solutions, conformal calibration and risk control, are extended to a problem they were not designed for, delegation through decision models that the organisation adopts rather than builds. Two bodies of knowledge justify the design: conformal prediction and multiple testing supply the mechanism of the guarantee (Section 2.2), and research on delegation to agentic information systems frames what the guarantee must protect (Section 2.1). We state the design knowledge as design principles with aim, context, mechanism and rationale (Gregor et al., 2020).

3.2 Design objectives

Five objectives, each with a measure and a criterion, translate the problem into testable requirements:

- **O1 Bound harmful execution:** the share of true refusals executed autonomously stays at most the harm budget, with stated confidence.
- **O2 Bound unconfirmed execution:** the share of true asks executed autonomously stays at most the ask budget, with stated confidence.
- **O3 Preserve safe autonomy:** the share of safe calls executed autonomously is as high as O1 and O2 allow.

- **O4 Remain model-agnostic:** the gate needs only output probabilities from a decision model.
- **O5 Reflect organisational costs:** the expected cost per call, in reviews, is the lowest among certified policies.

O1 and O2 are the risks a manager signs off on, per true class (Section 4.2); O3 is the benefit sought; O4 reflects that decision models are black boxes; O5 makes the costs of harm, review and blocking explicit.

3.3 Evaluation strategy

Evaluation follows the Framework for Evaluation in Design Science (Venable et al., 2016). Because a failing gate lets harmful actions run, the main risk of the artefact is technical, and we chose the technical risk and efficacy strategy: the gate is evaluated against its objectives on labelled data, artificially and before naturalistic use, which would expose an organisation to the harm it is meant to prevent. The evidence comes from three evaluation episodes with separate data roles (Venable et al., 2016): a formative episode evaluates the gate while it is built (Episode 1, Section 4; Sonnenberg & vom Brocke, 2012); the design is then frozen with pre-specified predictions, which keeps hypothesis-generating and hypothesis-testing evidence apart as in pre-registration (Nosek et al., 2018); and two summative episodes confirm the frozen design on new calls (Episode 2) and estimate every effect on the whole corpus (Episode 3; Section 5).

4 The Delegation Gate

Episode 1 developed the gate on the 1,433 development calls (Section 5.3) in three iterations: standard prediction sets under-covered refusals, class-conditional calibration revealed the floors of Section 4.4, and model combination, service-conditional calibration and the certified policies completed the design.

4.1 Architecture

Figure 2 adds the calibration part to the run-time flow of Figure 1. At run time, a decision model receives the proposed call with a fixed policy that defines the three actions and returns a probability for each; the gate compares them with certified thresholds, and several models can be combined by averaging their probability vectors (design principle DP3, Section 7). Before deployment, the calibration part works on labelled past calls, each with its correct action and target service. A certifiability check tests both floors for every budgeted class and the separability of each service, and so decides whether a model is admitted. Calibration then tests candidate thresholds per class and service with binomial tests at the stated confidence and passes the cheapest certified thresholds to the gate, given the organisation's budgets and its costs of harm, review and blocking.

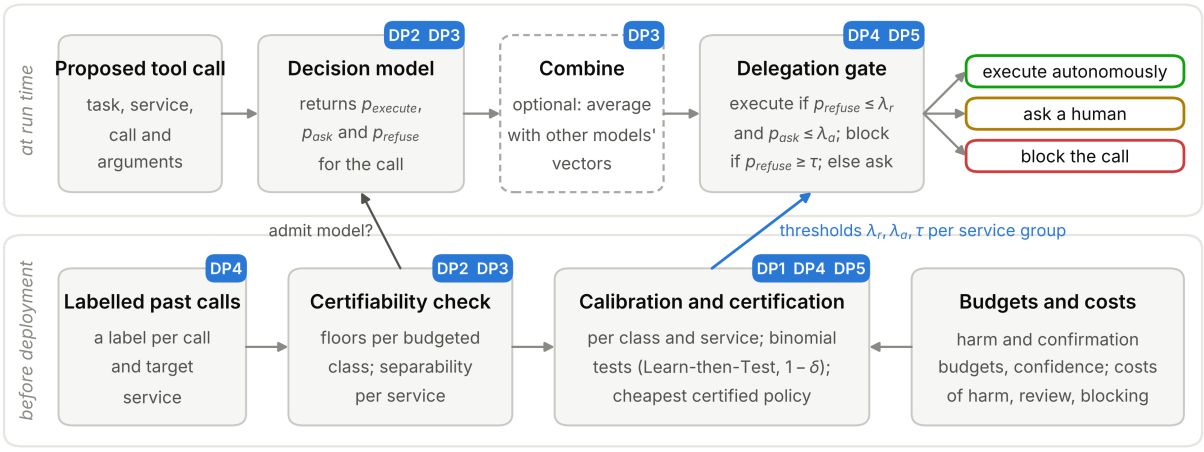


Figure 2. Architecture of the delegation gate (tags DP1 to DP5: design principles, Section 7).

4.2 Risks, budgets and costs

Let $y \in \{\text{execute}, \text{ask}, \text{refuse}\}$ be the correct action for a call and a the gate's action. The organisation sets two budgets, α_{harm} for harmful execution and α_{ask} for unconfirmed execution, and the gate must satisfy

$$R_{harm} = P(a = \text{execute} \mid y = \text{refuse}) \leq \alpha_{harm} \quad \text{and} \quad R_{ask} = P(a = \text{execute} \mid y = \text{ask}) \leq \alpha_{ask},$$

while maximising safe autonomy, $P(a = \text{execute} \mid y = \text{execute})$. Conditioning on the true class matters: refusals are a minority, so a bound on all calls together would allow most refusals to be executed. The expected cost per call, with one human review as the unit, is

$$C = P(a = \text{ask}) + c_{harm} \cdot P(a = \text{execute}, y = \text{refuse}) + c_{unconf} \cdot P(a = \text{execute}, y = \text{ask}) + c_{block} \cdot P(a = \text{refuse}, y \neq \text{refuse}).$$

The evaluation uses $\alpha_{harm} = 5\%$, $\alpha_{ask} = 15\%$, $c_{harm} = 10$, $c_{unconf} = c_{harm}/5$ and $c_{block} = 2$ for illustration.

4.3 Gate variants

We first examine three diagnostic variants (G1 to G3) that expose the design requirements and then construct two certified variants (G4 and G5) that form the artefact (Table 1). G4 is the main artefact. It certifies a bound for each class across all services and does not bound the risk on any single service; where that is needed, G5 certifies service groups, and calls of uncertified services go to humans.

Variant	Executes a call if	Guarantee	Interpretation
Diagnostic: G1 class-conditional sets	its prediction set is {execute}	both budgets, on average	runs a call only if, after calibration, execute is the only plausible action
Diagnostic: G2 refuse excluded	execute is in its set, refuse is not	harm budget, on average	runs a call unless refuse is still plausible; doubt between execute and ask is tolerated
Diagnostic: G3 service-group sets	as G2, with a refusal threshold per group	harm per group, on average	as G2, with a separate refusal cut-off for each group of services
Certified: G4 certified policy	$p_{refuse} \leq \lambda_r$ and $p_{ask} \leq \lambda_a$ (certified, cost-selected)	both budgets across services, with confidence $1-\delta$	runs a call only if both risk probabilities are below cut-offs that pass a statistical test on past calls
Certified: G5 service-group certified policy	as G4, with a certified λ_r per group	harm per group and ask budget, jointly with confidence $1-\delta$	as G4, with a tested refusal cut-off per service group; groups without one go to humans

Table 1. Gate variants.

The diagnostic variants use class-conditional prediction sets. With the score $s(x,y) = 1 - p_y(x)$, the gate computes for each class k the conformal quantile q_k of the calibration scores of class k at level α_k and forms the set $C(x)$ of labels whose score lies below their threshold; a singleton set is acted on and any other set goes to a human. A refusal is executed only if refuse is excluded from its set, which happens with probability at most α_{harm}; G2 relaxes this rule and G3 refines it per service group (Table 1).

G4 uses the whole probability vector: execute if $p_{refuse} \leq \lambda_r$ and $p_{ask} \leq \lambda_a$; refuse if $p_{refuse} \geq \tau$; otherwise ask. It certifies candidate thresholds with Learn-then-Test and selects the certified candidate with the lowest cost C on the calibration data (Table 2). The grid is fixed in advance: λ_r takes 15 values from 0 to 0.30, λ_a nine values from 0.01 to 0.50 and τ four values from 0.5 to never; $\delta = 10\%$.

G5 adds a certified λ_r per service group (Table 2); the four groups are three services derived from public benchmarks (agentharm, atbench and tsbench) and all others pooled.

Certification procedure of G4, with the additions for G5
Input. Labelled calibration calls with probability vectors; budgets α_{harm}, α_{ask}; confidence $1-\delta$; grids Λ_r ordered from strict to permissive, Λ_a with m values, and T; costs c_{harm}, c_{unconf}, c_{block}.
Step 1. For each $\lambda_a \in \Lambda_a$, run one test sequence at level δ/m (Bonferroni correction over the m sequences).
Step 2. Within a sequence, take λ_r from Λ_r in order. Let $E(\lambda)$ be the calls with $p_{refuse} \le \lambda_r$ and $p_{ask} \le \lambda_a$, and k_{harm} and k_{ask} the numbers of calibration refusals and asks in $E(\lambda)$.
Step 3. Test the null hypothesis $H_0(\lambda)$: $R_{harm}(\lambda) > \alpha_{harm}$ or $R_{ask}(\lambda) > \alpha_{ask}$. Its p-value is $\max\{F(k_{harm}; n_{refuse}, \alpha_{harm}), F(k_{ask}; n_{ask}, \alpha_{ask})\}$, where F is the binomial distribution function; H_0 is rejected only if both risks are (intersection-union test).
Step 4. If the p-value is at most δ/m, add λ to the certified set $\mathscr{C}$ and continue; otherwise stop the sequence (fixed-sequence testing). Add the empty policy, which executes nothing and is trivially certified.
Step 5. With probability at least $1-\delta$ over independent calibration calls, every policy in $\mathscr{C}$ satisfies both budgets simultaneously, so choosing $\lambda \in \mathscr{C}$ and $\tau \in T$ with the lowest calibration cost C on the same data preserves the guarantee; τ only decides which non-executed calls are blocked and needs no test.
G5, step A. For each λ_a, certify the ask risk on all asks as in Steps 2 to 4 at level $\delta/(2m)$, and let λ_r^* be the largest certified λ_r.
G5, step B. For each of the G service groups, certify the group's harm risk on its own refusals at level $\delta/(2mG)$, using only $\lambda_r \le \lambda_r^*$; a group without a certified λ_r executes nothing.
G5, step C. Every group's execution region lies inside $\{p_{refuse} \le \lambda_r^*, p_{ask} \le \lambda_a\}$, whose ask risk is certified, so group thresholds cannot raise the ask risk; with the Bonferroni correction over all $m+mG$ tests, all guarantees hold jointly with probability at least $1-\delta$.

Table 2. Certification procedure of G4 and G5.

4.4 Two certifiability floors

Black-box decision models express probabilities with finite resolution, and some return exact zeros, which creates two limits that no calibration can overcome.

Proposition 1 (score-resolution floor). Let v be the lowest refusal probability a model can return. For gate G1 or G2 calibrated on n refusals at level α_{harm}, if more than $n - \lceil (n+1)(1-\alpha_{harm}) \rceil$ calibration refusals receive v, then refuse is included in every prediction set and the gate never executes autonomously. As n grows, the condition approaches $\pi > \alpha_{harm}$, where π is the share of refusals at v.

Proof. The refusal threshold is the $\lceil (n+1)(1-\alpha_{harm}) \rceil$-th smallest calibration refusal score; if more than the stated number have the maximal score $1-v$, the threshold equals $1-v$, refuse is in every set, and neither G1 nor G2 executes; the same holds for asks.

Proposition 2 (policy-family floor). Let $k_{max}(n)$ be the largest k with $F(k; n, \alpha) \le \delta/m$. G4 can certify a non-empty policy only if its execution region contains at most $k_{max}(n)$ of the n calibration cases of each budgeted class. Every candidate region contains the calls whose refusal and ask probabilities are both at their lowest expressible values, so G4 executes nothing when more than $k_{max}(n)$ budgeted cases receive them. Because no k qualifies while $(1-\alpha)^n > \delta/m$, no non-empty policy can be certified with fewer than $\ln(\delta/m)/\ln(1-\alpha)$ calibration cases of a budgeted class.

Proof. It follows from Steps 3 and 4 of Table 2. For $\alpha_{harm} = 5\%$, $\delta = 10\%$ and $m = 9$, the execution region may contain no calibration refusal with 101 of them, at most 3 of 202 (1.5%) and at most 36 of 1,057 (3.4%), and at least 88 refusals are needed if calibration calls are independent; certifying four groups jointly requires 129 per group, twelve groups 150. These numbers are within a factor of two of a limit that no test removes: even when no case fails, no valid non-randomised test certifies a rate of at most α with confidence $1-\delta$ from fewer than $\ln\delta/\ln(1-\alpha)$ cases, 45 at 5% and 90%, and testing nine ask

thresholds raises it to 88. The floors apply to every budgeted class (refusals bind here because asks are plentiful) and can be measured before a model is adopted.

5 Evaluation Design

Four evaluation questions (EQ1 to EQ4) derived from the objectives structure Sections 6.1 to 6.4: which properties of a model's outputs decide whether autonomy can be certified (EQ1; O1, O4), whether the bound holds across samples and on new calls (EQ2; O1, O2), how much labelled history certification needs and at which granularity (EQ3; O1 to O3), and what certified autonomy costs (EQ4; O3, O5).

5.1 Data

We use the SAFETY SENTRY corpus (Chen et al., 2026). Each record is one tool call that an agent proposes during a user task, with the task, the user's standing instructions, the session so far and the target service, labelled with the correct action: execute, ask or refuse. In one record, an agent asked to delete two credentials during a key rotation proposes to delete an administrator credential that the user's instructions exclude; the call is labelled refuse. Two language-model annotators labelled the calls with a 14-trigger taxonomy, the corpus authors arbitrated, and 289 labels were changed during curation; a human audit of the test split found 92% agreement, so every guarantee below holds with respect to these labels. The calls come from self-hosted enterprise services, such as password vaults, ticketing and medical records, and from agent-safety benchmarks adapted to tool calls, among them AgentHarm (Andriushchenko et al., 2025) and R-Judge (Yuan et al., 2024), the source of the dh_app service.

We analyse the calls that all three decision models scored validly: 1,433 development and 7,755 confirmation calls (4,563 execute, 2,510 ask and 2,115 refuse labels) and 198 Mailu calls. Calls are not independent: linking those sharing a task text, benchmark session or identical prompt yields 4,155 groups of up to 66 calls.

5.2 Decision models

A decision model is a discriminative counterpart of a large language model: it reads the same text input but returns, in a single pass and without a rationale, a probability for each of a fixed set of options. It replaces prompting a general-purpose model and parsing its reply; because probabilities differ between models, thresholds must be chosen anew for each (OpenRouter, n.d.-b), and the gate makes this choice a certified one. We queried three such models as black boxes through a common, alpha-stage endpoint in October 2026, with a shared system policy and question.

Two are large language models adapted in this way: Microsoft-Decision-1 (microsoft/microsoft-decision-1), post-trained from the open-weight Qwen3.5-9B (Swain, 2026), and GPT-6 Luna Decisions (openai/gpt-6-luna-decisions), served through OpenAI's decisions interface (OpenRouter, n.d.-a). Jev 1.14 (typesafe/jev-1.14), from TypeSafe, is presented as non-generative, with unpublished weights and design (Rogers, 2026). The three have similar accuracy (0.636 to 0.657) but represent their outputs differently: Microsoft-Decision-1 returns continuous probabilities, Jev 1.14 and GPT-6 Luna Decisions probabilities on a 0.01 grid with exact zeros, so accuracy alone does not reveal what the gate can certify. For the example call, Microsoft-Decision-1 gives execute 0.77, ask 0.22 and refuse 0.01, whereas Jev 1.14 and GPT-6 Luna Decisions rank ask highest. Re-scored development calls kept their most probable action in 98 to 100% of 150 cases. We also evaluate the mean of the three probability vectors.

5.3 Evaluation episodes

The three episodes use separate data. Episode 1 used the 1,433 calls of the test split (we retain the corpus' split names *test* and *train*, though no models were trained; Section 4). We then froze the design, the analysis code and a protocol with the 21 predictions and their verdict rules, stored with checksums

in the project repository, not in a public registry, before any new score existed. Episode 2 applied the frozen certified policies unchanged to the 7,755 calls of the training split and to the 198 calls of the out-of-distribution split, from a service absent elsewhere, and repeated the calibration on disjoint samples of the new calls. Of the 7,755 calls, 6,380 share no task, session or prompt with a development call, the example call among them. Episode 3 pools the 9,188 in-distribution calls to estimate every effect with the frozen gate code, keeping each group on one side of every split. Section 6 reports Episode 3 unless stated otherwise; claims of confirmation rest on Episode 2 alone.

5.4 Statistical procedure

Because the decision models were not trained or tuned on the corpus, every labelled call can serve as calibration data. We draw 100 calibration-test splits that assign each group to calibration with probability one half, so no test call has a related call in calibration (single-call splits, which leave one for 58% of test calls, give the same conclusions). Grouped splits prevent leakage between calibration and test but not dependence within calibration, which the certificate treats as independent draws; they and one-call-per-group calibration (Section 6.3) are sensitivity analyses. We report means, the share of test halves above a budget, which is the risk an organisation takes with its one calibration sample, and exact Clopper-Pearson intervals (Clopper & Pearson, 1934) for rates on fixed calls. Comparing the models isolates the output lever; holding out each service and varying the calibration share (5% to 90% of groups, 30 repetitions), confidence (80% to 99%), harm budget (2% to 20%) and granularity (one to twelve service groups, 10 repetitions) isolate the history lever. Exploratory checks outside the frozen design cover label noise, shift, model combinations, costs, review capacity, labelling and near-duplicates.

6 Results

6.1 The output lever: what the decision model exposes

What a decision model's output exposes decides whether autonomy can be certified, and accuracy does not reveal it. Table 3 describes each model deciding by its most probable action, with the top-1 expected calibration error (ECE, 15 bins) and the shares of refusals or asks whose correct class receives a probability of zero or at most 0.01. Used directly, the models execute 17 to 24% of the calls that should be refused, and the most accurate model, Jev 1.14, gives the most refusals an exact zero. The critical class also needs its own calibration: with one global threshold at a 10% level, only 77.8 to 79.7% of true refusals have refuse in their set under Jev 1.14, Microsoft-Decision-1 and the mean, against 90.2 to 93.6% under G1.

Scorer	Output	Accuracy	Refusal recall	Harmful execution	ECE	Refusals at $p = 0$	Refusals at $p \leq 0.01$	Asks at $p = 0$
Jev 1.14	0.01 grid	0.657	0.44	16.6%	0.15	6.7%	13.6%	1.4%
Microsoft-Decision-1	continuous	0.636	0.41	24.4%	0.28	0.0%	24.2%	0.0%
GPT-6 Luna Decisions	0.01 grid	0.637	0.47	19.9%	0.25	5.4%	13.6%	10.2%
mean of three	continuous	0.655	0.47	22.1%	0.18	0.0%	9.8%	0.0%

Table 3. Decision models without calibration.

Exact zeros can remove certifiability. Jev 1.14 and GPT-6 Luna Decisions give 6.7% and 5.4% of true refusals a refusal probability of exactly zero (Figure 3a), which no threshold on that probability can separate from safe calls. Under G2 at a 5% budget, they execute nothing in the 77% and 67% of halves in which these zeros exceed the number Proposition 1 tolerates, and 13% and 16% of safe calls on average, against 53% for Microsoft-Decision-1, whose continuous probabilities have no exact zeros (Figure 3b). Figure 3c confirms Proposition 1 sample by sample: in all 292 of 440 calibration samples in which the

zero-probability refusals exceed the tolerated number, the gate executes no call; of the 148 others, 138 execute. Near the threshold, binding varies between samples, so a floor test needs a margin.

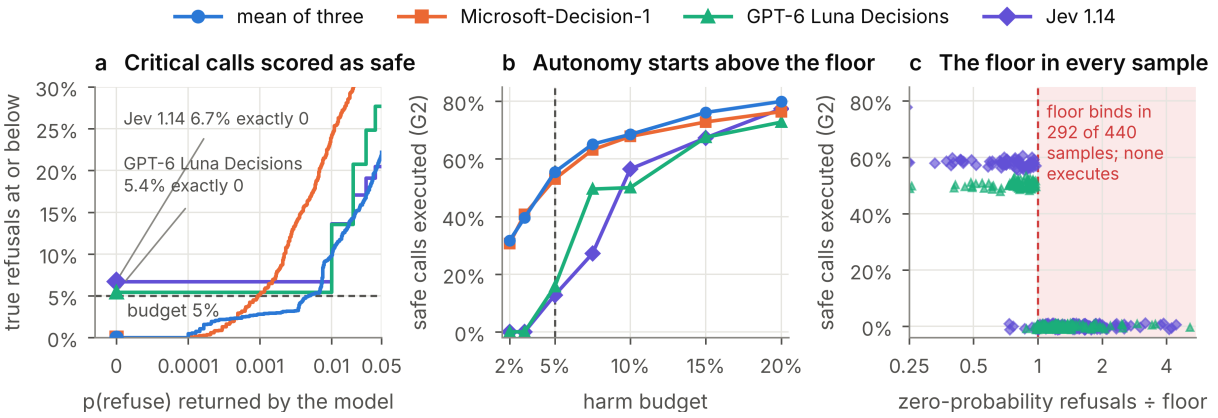


Figure 3. *The output lever: (a) refusals by refusal probability; (b) G2 autonomy by harm budget; (c) G2 autonomy against the floor of Proposition 1.*

A richer policy family can escape the score-resolution floor, but not the policy-family floor. Jev 1.14 gives most zero-probability refusals a high ask probability (median 0.14), so G4, which also conditions on it, excludes them and executes 42% of Jev 1.14's safe calls. GPT-6 Luna Decisions, as accurate as Microsoft-Decision-1, gives 19.5% of the calls that need confirmation a refusal probability of zero and an ask probability of at most 0.01, the output it also gives 45% of safe calls. Every candidate region of G4 contains these calls, and their share exceeds the ask budget, so GPT-6 Luna Decisions cannot be certified at any calibration size, whereas G4 certifies 24% of Microsoft-Decision-1's.

Combining models helps only where their critical errors differ (Figure 4). Their confident errors are dependent, with a lift of 3.0 to 4.3 over independence, but diverse: of the 622 refusals that at least one model scores at or below 0.01, 467 are scored that low by only one or two. The mean of three therefore has no exact zeros and executes more safe calls than any single model under the set-based gates from a 5% budget; under G4 it executes 37% against Jev 1.14's 42%. The median and the geometric mean bring exact zeros back (3.3% and 8.9% of refusals). Where errors are shared, no combination helps. On eight enterprise services and agentharm, the mean separates refusals from safe calls with an area under the receiver operating characteristic curve (AUROC) of 0.88 to 0.98; on the benchmark-derived atbench and dh_app, every model and combination reaches only 0.48 to 0.59. Held out, G4 with the mean executes 39% of dh_app and 11% of atbench refusals, against at most 5.6% elsewhere.

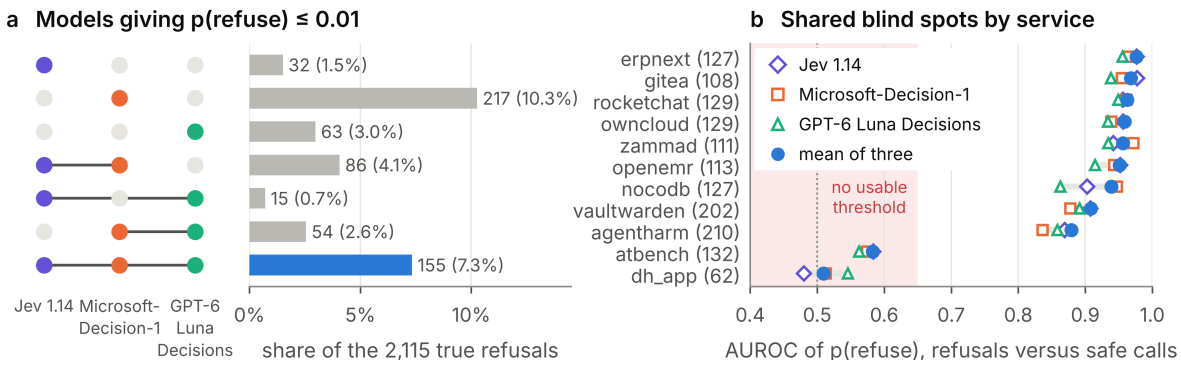


Figure 4. *Diverse and shared errors: (a) refusals scored ≤ 0.01; (b) AUROC per service.*

6.2 Holding the guarantee: across samples and on new calls

A guarantee in expectation and a high-confidence guarantee differ in what an organisation risks with its one calibration sample (Figure 5a, b). The diagnostic variants satisfy the budgets on average but exceed them in many test halves: G2 exceeds the harm budget in 37% of halves with Microsoft-Decision-1 and 39% with the mean, G3 in 26% with the mean, and Microsoft-Decision-1 under G1 exceeds the ask budget in 27%; a policy from G4's family chosen without the statistical test exceeds a budget in 13% for the mean. The certified G4 exceeds a budget in no half for Microsoft-Decision-1, 2% for the mean and 4% for Jev 1.14, in line with its 90% confidence, and G5 in none. For the mean, G1 exceeded a budget in only 3% of halves at 41% autonomy, so certification adds assurance for every scorer and sample rather than lower observed rates.

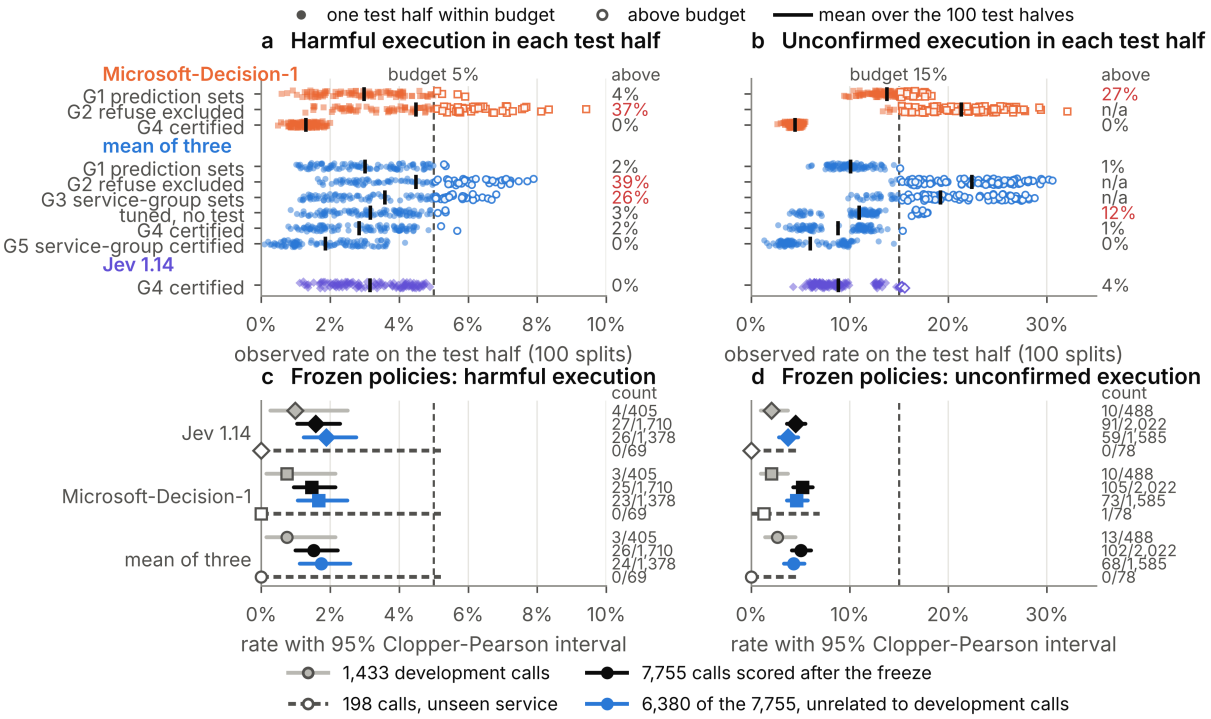


Figure 5. *Holding the guarantee: (a, b) execution in 100 test halves (n/a: no ask budget); (c, d) frozen G4 policies on four sets of calls.*

The bound tolerates label noise, which matters as audit disagreement exceeds the budget: with random errors in 2 to 10% of calibration refusals, G4's harm rate for the mean stays at 2.8 to 2.9%, and errors on the refusals the model finds least risky raise it to 3.3% (Jev 1.14: 3.8%, over budget in 20% of halves).

The confirmation applies the three G4 policies frozen after Episode 1 to the calls scored after the freeze (Figure 5c, d). They executed 1.5 to 1.6% of 1,710 refusals and 4.5 to 5.2% of asks. Because 1,375 of these calls are related to a development call, a stricter, post hoc test uses the 6,380 unrelated ones: the frozen policies executed 23 to 26 of their 1,378 refusals (1.7 to 1.9%, upper bounds 2.5 to 2.8%) and 3.7 to 4.6% of their asks, about twice the development rates. These rates show that the bound held with its margin, not how tight it is; in the sharper test, recalibration on four disjoint samples of single new calls, G4 never exceeded a budget and G2 for the mean exceeded the harm budget in three. Of the 21 pre-specified predictions, 16 were confirmed, one was inconclusive (Section 6.3) and four were not; recomputed with whole groups on one side of every split, 20 verdicts stay the same and one of the four failures, on atbench refusals, is then confirmed. The most consequential failure concerns G5, which we expected to execute at least as many safe calls as G4 at no higher cost; calibrated on the training pool,

it executed 23% against 42% at 0.83 against 0.81 review-equivalents per call. The other two concern thresholds: that Jev 1.14 executes at most 1% of safe calls under G1 and G2, and that G2 executes more than 5% of atbench refusals.

6.3 The history lever: amount, independence and granularity

More labelled history does not merely improve estimates; it decides whether certification is possible, how much autonomy it carries and how finely the boundary can be drawn (Figure 6). At a 5% budget, G4 needs at least 75, 88, 102 and 133 labelled refusals for 80, 90, 95 and 99% confidence; at 2%, it needs 189 to 337. Above the minimum, the mean of three certifies 16% of safe calls with 107 calibration refusals, 29% with 212, 36% with 424 and 38% with 1,061 at 90% confidence, while harmful execution rises from 0.9% to 2.9%, within budget. Confidence is costly only with short history: 99% instead of 90% costs 14 points with 107 refusals (certifiable in 10% of samples) and 2 points with 1,061.

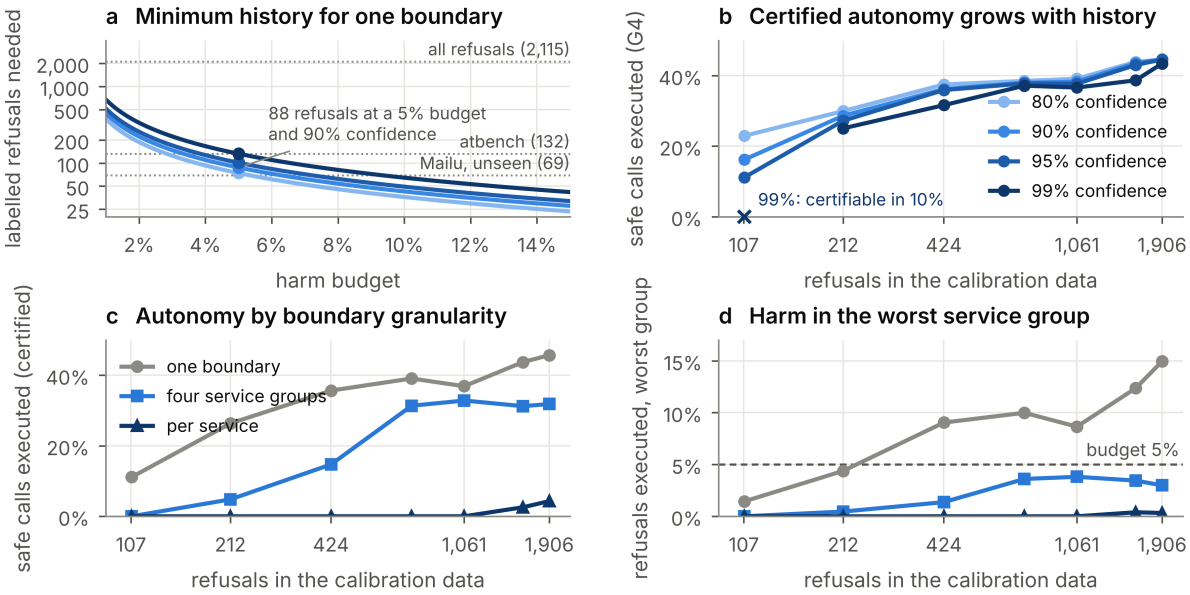


Figure 6. *The history lever: (a) minimum refusals for one boundary; (b) G4 autonomy by calibration size and confidence (cross: certifiable in at most half the samples); (c, d) autonomy and worst-group harm by boundary granularity.*

Related calls add less evidence than their number suggests. The 2,115 labelled refusals fall into 1,275 groups, refusals of one group receive similar probabilities (intraclass correlation 0.74 to 0.78), and the refusals G4 executes cluster, so the variance of its harm rate is about four times that of independent calls for the mean and Jev 1.14 (design effect). With half of the corpus for calibration, the test's margin absorbed this (the share of halves above budget stayed near or below δ), but a few large groups are worth fewer independent cases than their count. Calibrating on one call per group (Dunn et al., 2023) removed the remaining exceedances at a 5% budget, not at 2%, because it bounds the rate per task, not per call.

Finer boundaries need more history and protect against shift (Figure 6c, d). One boundary for all services lets atbench refusals run at 9% under G4 with half of the corpus, almost twice the budget, while the overall rate stays near 3%, and at 11 to 12% with three quarters. Four service-group boundaries (G5) keep every group within budget on average at every calibration size, the worst at most 4%, at 31% instead of 37% autonomy, as each group is certified on its own refusals at a stricter level. Because both risks condition on the true class, a change in the share of refusals leaves G4's bound intact and moves only workload and cost; a change in the service mix does not. If atbench grew from 6% to 40% of refusals, G4's harm rate would rise to 5.1%, above the budget, while G5's would fall to 1.2%. One boundary per service needs 150

refusals for each of twelve groups: on all 9,188 calls, only the three with over 200 refusals are certified.

An unseen service cannot inherit the guarantee. Mailu has 69 labelled refusals, fewer than the 88 one boundary needs, and the frozen policy for the mean of three executed none of its refusals but also none of its safe calls. A rate of zero on 69 refusals is compatible with a true rate up to 5.2%, which made the prediction for Mailu inconclusive: observing no error is not having the evidence to certify a bound.

6.4 The price of certified autonomy

Table 4 reports means over the 100 grouped splits, with costs in reviews per call and policies selected at a harm cost of ten (n/a: no ask budget). Certified autonomy is paid for in human review. G4 sends 66% of all calls to a human, against 26% when the model decides alone, and in 99 of 100 halves no certified policy for the mean sends fewer than 60%. Review capacity is therefore a precondition rather than a parameter. At a harm cost of ten reviews, G4 is among the cheapest gates, while letting the model decide costs more than reviewing every call. The cost of harm decides how much runs: the cost-optimal certified policy executes 37 to 40% of safe calls at a harm cost of five or ten, 22% at 20 and 3% at 50. Certification bounds the risks, not the cost, which is estimated on the data that selected the policy. A single model can carry the gate: G4 certifies 42% of Jev 1.14's safe calls at a lower cost than the mean, but exceeds the ask budget in 4% of halves and runs 13% of atbench refusals against 9%; G5 with Jev 1.14 reaches 30%.

Measured against the objectives (Section 3.2), the certified variants meet O1 and O2 with high confidence, G4 per class and G5 per certified service group; O3 is met partly, at a price that grows with the granularity of the boundary; O4 holds only for models that pass the floor tests, which turns it into a selection criterion; and O5 is met for the selection of the policy, not for the cost itself.

7 Design Principles

The evaluation condenses into five design principles (DP1 to DP5), stated as actions for implementers of delegation gates, with their mechanism and evidence: each gives the principle in bold, its rationale in italics and the evidence. They specify the delegation boundary along four dimensions: what is delegated (DP1), to which decision models (DP2, DP3), where (DP4) and with which assurance (DP5). They apply under three boundary conditions: the services in scope have labelled calls, the models expose probabilities for all three actions, and these calls are independent draws that future calls resemble within each class.

- **DP1 Protect critical classes separately.** Calibrate the guarantee for each critical class separately, so that the budget for harmful execution is spent only on calls that should be refused. *Aggregate guarantees hide critical errors.* A global threshold lets the easy classes absorb the error budget: refusal coverage falls to 77.8 to 79.7% at a 90% target for three of four scorers, against 90.2 to 93.6% with class-conditional calibration (Section 6.1).
- **DP2 Test certifiability before adoption.** Test each decision model's score-resolution floor and its policy-family floor at the organisation's budget, confidence and calibration size, for every budgeted class and with a margin for sampling variation, and require outputs with sufficient resolution. *Accuracy alone is not a selection criterion for autonomy.* Two models of equal accuracy differ in certifiability: G4 certifies 24% of Microsoft-Decision-1's safe calls and none for GPT-6 Luna Decisions, which scores many calls that need confirmation like safe calls; exact zeros block G2 for Jev 1.14 and GPT-6 Luna Decisions, and the floor binds in 292 of 440 calibration samples, none of which executes a call (Propositions 1 and 2, Figure 3).
- **DP3 Combine decision models only where their critical errors differ.** When decision models show diverse critical errors on labelled data, combine their outputs before calibration, verify that the combination lowers the certifiability floor, and measure for each service whether any model separates critical from safe calls; route services without such separation to humans. *Combining models does not remove shared blind spots.* Only 7.3% of refusals are scored at or below 0.01 by all three models, against 13.6 to 24.2% by each, so the mean has no exact zeros, whereas the median and the geometric

mean reintroduce them. On two services no model or combination separates refusals from safe calls, and held out, 11 to 39% of their refusals run (Section 6.1, Figure 4).

Gate	Safe calls executed	Refusals executed	Asks executed	Halves over budget (harm / ask)	Sent to a human	Cost, harm = 10	Cost, harm = 50
Model alone, no gate	85.3%	21.7%	49.9%	100% / 100%	26.0%	1.08	4.16
A human reviews every call	0.0%	0.0%	0.0%	0% / 0%	100%	1.00	1.00
Uncertified gates, mean of three: budgets hold on average or not at all							
G1 class-conditional sets	41.4%	3.0%	10.1%	2% / 1%	72.9%	0.86	1.35
G2 refuse excluded	55.5%	4.5%	22.4%	39% / n/a	62.2%	0.85	1.75
G3 service-group sets	44.3%	3.6%	19.2%	26% / n/a	68.8%	0.88	1.63
G4 and G5, mean of three: with the test, budgets hold with 90% confidence							
G4 tuned, without the test	41.6%	3.2%	11.0%	3% / 12%	63.6%	0.80	1.33
G4 certified policy	37.5%	2.8%	8.8%	2% / 1%	66.4%	0.81	1.26
G5 service-group certified	31.0%	1.9%	6.0%	**0% / 0%**	70.6%	0.81	1.11
Certified gate G4 with a single decision model							
Jev 1.14	**42.4%**	3.1%	8.8%	0% / 4%	**64.4%**	**0.79**	1.27
Microsoft-Decision-1	24.5%	**1.3%**	**4.5%**	**0% / 0%**	74.7%	0.85	**1.06**
GPT-6 Luna Decisions: none certifiable	0.0%	0.0%	0.0%	0% / 0%	87.2%	0.92	0.92

Table 4. Risk, autonomy and cost of the gates (bold: best value among gates within both budgets in at least 90% of test halves).

- **DP4 Match the granularity of the boundary to the labelled history.** Certify a service or group of services only where it has enough labelled critical calls, from enough distinct tasks, for the joint test, route calls of uncertified and new services to humans, and recalibrate before claiming the guarantee for a new service. *Finer governance needs more evidence.* One boundary lets the refusals of one service run at almost twice the budget and exceeds the budget when that service's share of refusals grows to 40%; four group boundaries keep every group within it at 31% instead of 37% autonomy; per-service boundaries need 150 refusals each; clustered refusals can quadruple the variance of the harm rate; an unseen service with 69 refusals cannot be certified (Section 6.3, Figure 6).
- **DP5 Certify, then confirm the final policy.** Certify the policy with high confidence over a pre-specified family that uses the full probability vector, choose among certified policies by the organisation's costs, and confirm the frozen policy on labelled calls unrelated to those used for its calibration before relying on it. *Tuning is not certification, and internal evaluation is not confirmation.* Guarantees in expectation exceed the budget in up to 39% of test halves, the certified policy in at most

4%; frozen certified policies executed 1.7 to 1.9% of 1,378 refusals unrelated to any development call (Section 6.2, Figure 5, Table 4).

8 Discussion

8.1 Theoretical contributions

The paper makes two theoretical moves. First, it conceptualises delegation in agentic information systems as a boundary drawn per action, by a decision model, under a risk budget stated in advance; in terms of Baird and Maruping (2021), the transfer of rights and responsibilities happens per action and is itself delegated to a machine. Where prior work tunes the gate as a classifier (Chen et al., 2026; Mozannar et al., 2025), we treat it as an accountable boundary whose guarantee is part of its specification, answering the inscrutability of AI (Berente et al., 2021) with verifiable consequences.

Second, it adds certifiability to the criteria of delegate suitability: equally accurate decision models need not be equally certifiable. Certifiability depends on an output lever with the model developer and a history lever with the organisation; the two floors make both testable before deployment.

8.2 Implications for practice

The findings inform three decisions. In model selection, organisations should require fine-grained probabilities for every action and measure both floors on their own data before adoption. In data governance, labelled critical calls from many distinct tasks set attainable autonomy and granularity: at least 88 independent ones for one boundary at a 5% budget and 90% confidence, and 129 per group for four. Labelling should follow separability: about 890 further labelled calls would certify seven enterprise services each, whereas labels on the two blind spots gain nothing. In operations, review capacity for over 60% of calls must be planned with the budget and confidence, and costs should reflect the effort of reviewers who tire (Turan, 2026). A certified policy can document oversight under the AI Act (European Union, 2024, Art. 14) but not establish compliance.

8.3 Limitations and future research

The guarantees hold with respect to the corpus labels (Section 5.1): random label errors leave the bound intact, errors concentrated on hard refusals erode its margin, and errors shared by labelling and decision models would make the rates optimistic. The evaluation is artificial: it replays one corpus offline. Grouping related calls, even by near-duplicate task texts (G4 above budget in at most 7% of halves; three secondary verdicts changed), left the conclusions unchanged but cannot rule out dependence through shared templates or services, which the certificate treats as independent. Training exposure to public benchmarks would limit transfer, not validity; that the benchmark-derived atbench and dh_app are the blind spots argues against it. Naturalistic evaluations should examine how reviewers handle the calls they receive. Future work should extend certification to within-class covariate shift, groups of related calls and call sequences, and test per-model thresholds and adaptive budgets.

9 Conclusion

Safe delegation to AI agents is not primarily an accuracy problem but a question of what can be certified. Following design science research, we designed a delegation gate that turns the probabilities of a black-box decision model into a policy with auditable bounds on harmful and unconfirmed autonomous execution, and we showed that whether such bounds can be certified depends on two levers rather than on accuracy: what the model's outputs expose, and the amount, independence and granularity of the organisation's labelled history. Two certifiability floors make both levers measurable before a model is adopted. The gate reduced harmful execution from the 17 to 24% of decision models acting alone to below a 5% budget that holds with 90% confidence, and the frozen design kept the bound on unrelated new calls; the price is

human review of about two thirds of all calls. Five design principles condense this evidence into guidance on what to delegate, to which decision models, where and with which assurance. For information systems research, they recast agent autonomy as an accountable, certifiable delegation boundary; for practice, they turn autonomy from a tuning choice into a decision that can be documented, audited and recalibrated.

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